ECE 2305 – Introduction to C Programming
Tutorial 06
Arithmetic Expressions

Write a C++ application that will declare the following variables (x, y, and z) with initial values of 10.0 and will execute the following formulas. All variables are to be double-precision floating point numbers.

$$a = 3x + y$$

$$b = \frac{x + y}{7}$$

$$c = \frac{3x + y}{z + 2}$$

Input

```
// ECE 2305 Tutorial 05 - Arithmetic Operators and Order of Precedence
#include <iostream>
using namespace std;

int main()
{      cout << "Arithmetic Operators and Orders of Precedence" << endl;
       cout << endl;

       cout << "x = y = z = 10.0" << endl;
       cout << endl;
       double x = 10.0;
       double y = 10.0;
       double z = 10.0;
       double a = 3 * x + y;
       double b = (x + y) / 7;
       double c = (3 * x + y) / (z + 2);
       cout << "a = 3 * x + y                ---> a = " << a << endl;
       cout << "b = (x + y) / 7              ---> b = " << b << endl;
       cout << "c = (3 * x + y) / (z + 2)    ---> c = " << c << endl;
       cout << endl;

       system("pause");
       return 0; }
```

Output

```
C:\Users\anmal\Desktop\ECE
Arithmetic Operators and Orders of Precedence

x = y = z = 10.0

a = 3 * x + y                  ---> a = 40
b = (x + y) / 7                ---> b = 2.85714
c = (3 * x + y) / (z + 2)      ---> c = 3.33333

Press any key to continue . . .
```